

A Simplified Replication of Rust's Dynamic Bus Engine Replacement Model

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Core question

When should a bus operator keep using an old engine, and when should the engine be replaced?

Dynamic trade-off

- Replacing is costly today.
- Keeping an older engine raises current and future maintenance costs.
- Mileage is the key state variable.

Why it matters

- Rust (1987) is a canonical dynamic discrete-choice model.
- The project connects theory, computation, estimation, and counterfactual simulation.

Team expertise level and what we learned

Starting point

- Strong economic intuition about optimal replacement.
- Limited prior experience implementing a full dynamic discrete-choice estimator from scratch.
- Needed to translate a structural model into a computable algorithm.

What we learned

- How to solve a Bellman fixed point numerically.
- How NFXP nests dynamic programming inside likelihood maximization.
- How value functions, hazards, stationary distributions, and counterfactual demand are connected.
- Why optimization diagnostics matter in structural estimation.

Main gain: we moved from “writing down” a dynamic model to actually estimating and using it.

Techniques and tools used

Model ingredients

- State: mileage bin $x \in \{0, \dots, 89\}$, each bin = 5,000 miles.
- Choice: keep ($i = 0$) or replace ($i = 1$).
- Cost: linear maintenance cost $c(x) = \theta x$.
- Shocks: Type-I extreme value, giving dynamic logit probabilities.
- Discount factor fixed at $\beta = 0.9999$.

Computational tools

- Python implementation from scratch.
- NumPy/Pandas for data and matrix operations.
- SciPy optimization using L-BFGS-B.
- Matplotlib for hazards, value functions, distributions, and demand curves.
- Warm starts and fallback nonlinear root solving for convergence.



Full model and NFXP estimation method

Model primitives

$$\underbrace{x_t \in \{0, \dots, 89\}}_{\text{States}}, \quad \underbrace{i_t \in \{0, 1\}}_{\text{Choice}}, \quad \underbrace{\Theta = (RC, \theta, p_0, p_1, p_2)}_{\text{Parameters}},$$
$$\underbrace{u_0(x) = -\theta x}_{\text{Maintenance Cost}}, \quad \underbrace{u_1(x) = -RC}_{\text{Replacement Cost}}, \quad U_j(x) = u_j(x) + \varepsilon_j.$$

Mileage transitions

$$P_0(y \mid x) = \sum_{d=0}^2 p_d \mathbf{1}\{y = \min(x + d, 89)\},$$
$$P_1(y \mid x) = \sum_{d=0}^2 p_d \mathbf{1}\{y = \min(d, 89)\}.$$

Bellman equation and policy

$$W(x) = \log\{\exp(v_0(x)) + \exp(v_1(x))\},$$
$$u_j(x) = u_j(x) + \beta \sum_y P_j(y \mid x) W(y), \quad \beta = 0.9999,$$

Likelihood in the outer loop

$$q_1(x; \Theta) = \Pr(i = 1 \mid x; \Theta) = \frac{\exp(v_1(x))}{\exp(v_0(x)) + \exp(v_1(x))}.$$
$$\ell(\Theta) = \sum_t \left[i_t \log q_1(x_t; \Theta) + (1 - i_t) \log q_0(x_t; \Theta) \right]$$
$$+ \sum_{d=0}^2 n_d \log p_d, \quad q_0(x; \Theta) = 1 - q_1(x; \Theta).$$

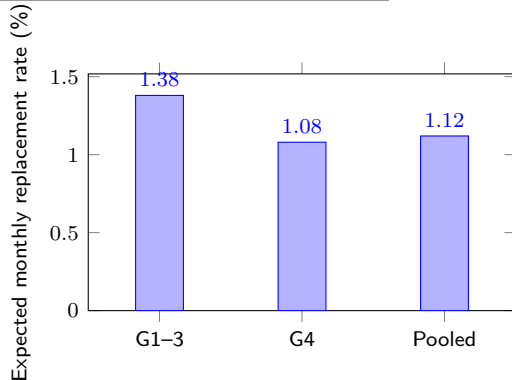
Nested Fixed Point algorithm

- 1 *Outer loop*: propose Θ , impose constraints, and build P_0, P_1 .
- 2 *Inner loop*: solve $W = T_\Theta(W)$ by value iteration; use a root solver if value iteration fails.
- 3 Compute q_0, q_1 , evaluate $\ell(\Theta)$, and update Θ with L-BFGS-B until convergence.

Main estimation results

Sample	RC	θ	p_0	p_1	p_2
Groups 1–3 (newer buses)	26.27	0.0127	0.3149	0.6793	0.0058
Group 4 (older buses)	66.01	0.0247	0.4059	0.5847	0.0094
Pooled	13.17	0.0035	0.3286	0.6675	0.0039

- Group 4 (older buses) has much larger estimated replacement cost and a steeper maintenance-cost slope.
- The pooled model flattens group differences and changes long-run predictions.
- Most monthly mileage changes are zero or one 5,000-mile bin.



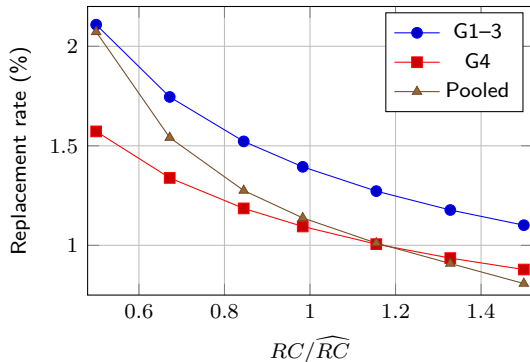
Challenges and outcomes

Challenges overcome

- NFXP is demanding: each likelihood evaluation solves a dynamic program.
- Fixed-point and optimizer convergence required careful monitoring.
- The heterogeneity diagnostic flagged a likely local-convergence or likelihood-comparability issue.

Final outcomes

- Replacement hazards rise with mileage.
- Higher replacement costs reduce equilibrium replacement demand.
- Pooling heterogeneous groups changes counterfactual predictions.



Cool part: a micro replacement decision generates a model-implied fleet-level demand curve.